ETERNAL RACING

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

What is Moksha?

Moksha is IIT Gandhinagar's Motorsports club that designs, builds, and races electric all-terrain vehicles in SAEINDIA eBAJA.

What is BAJA SAE?

SAEINDIA is India's automotive engineering society, and BAJA SAEINDIA is its flagship national collegiate competition

NATRAX, Pithampur, Dist Dhar, Madhya Pradesh

Our Journey

2024

2025

Now

2nd in Virtual Round eBAJA 2025 in Gujarat

28th in Virtual Round eBAJA 2025 overall

Our Journey

2024

2025

Now

in Virtual Round eBAJA 2026 in Gujarat

1st

in Virtual Round eBAJA 2026 overall

18th

How we build our eBAJA vehicle

Research & DFMEA

→

SOLIDWORKS

Ansys

CAD design, assembly & CAE analysis

→

Manufacturing & Assembly

→

Testing & Rulebook Validation

Our Vision

Improve Baja rankings

Foster motorsports culture

Formula student prospects

Team Structure

Prof. Chelva
Faculty Advisor

Prof. Madhu
Faculty Advisor

Pratham Wala
Vice-Captain &
Brakes Lead

Aditya Yadav
Captain &
Suspension Lead

Rishi Pinara
Vice-Captain &
Sponsorship Lead

Viraat Krishna
Electrical
Powertrain Lead

Pradnyesh Phadnis
Chassis & Marketing
Lead

Khushal Manani
Finance, Procurement
& CAE Lead

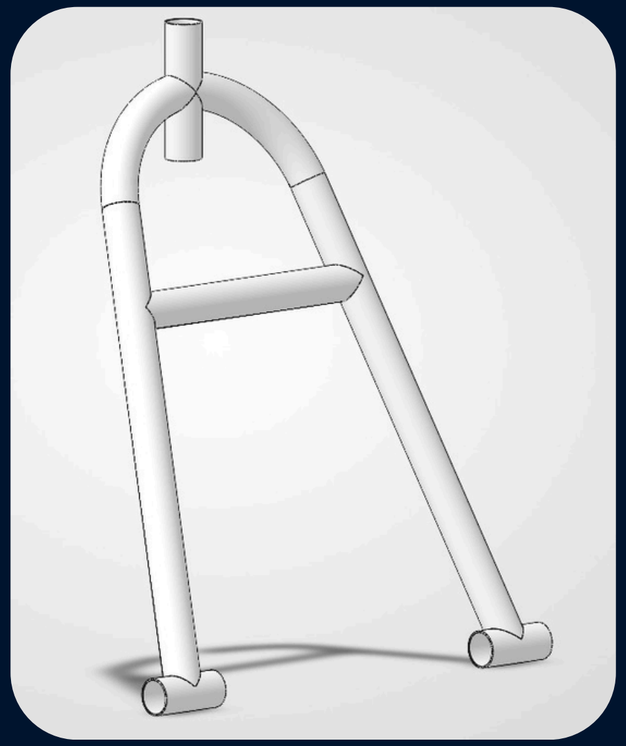
Kush Mehta
Mechanical
Powertrain Lead

Dhruv Meshram

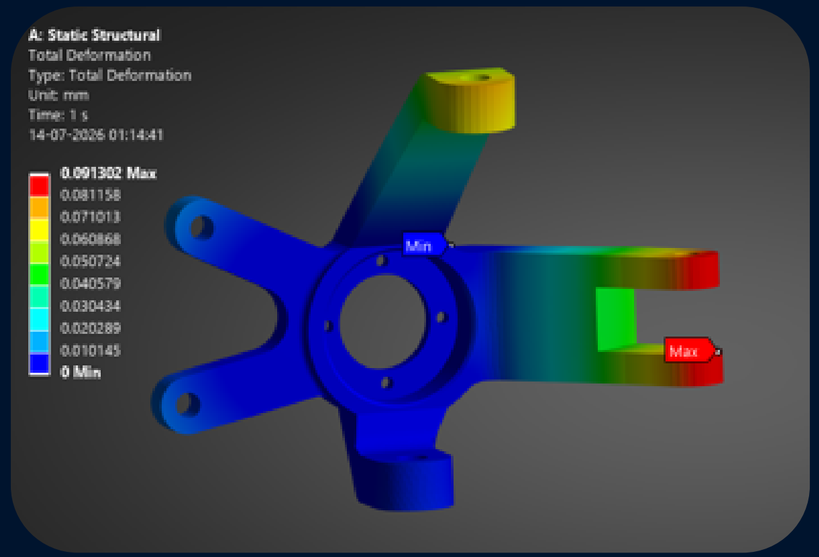
Harsh Kumar
Steering Lead

Suspension

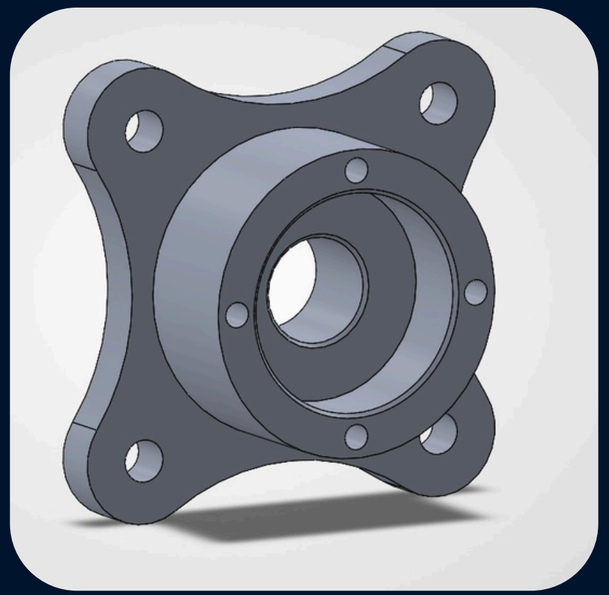
Designs and optimizes the suspension geometry for stability, ride comfort, and off-road performance



Right Front Lower Control Arm



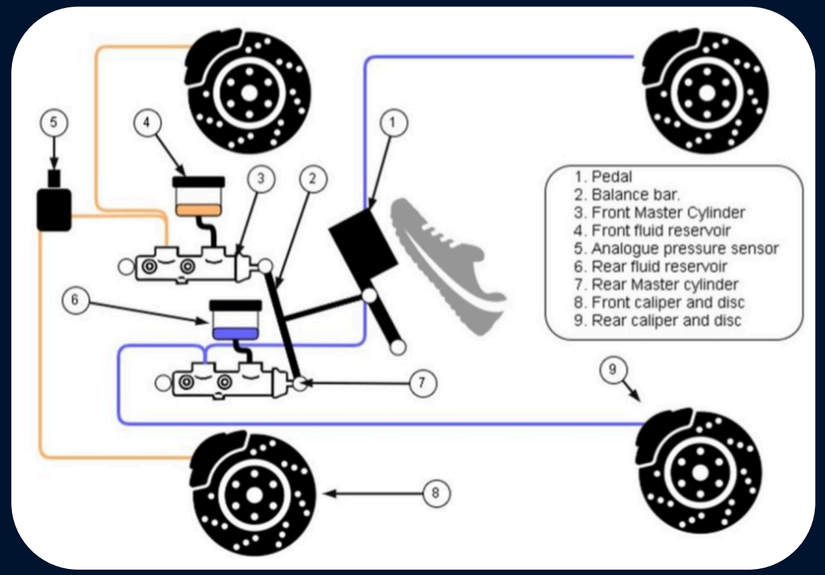
Right Front Knuckle



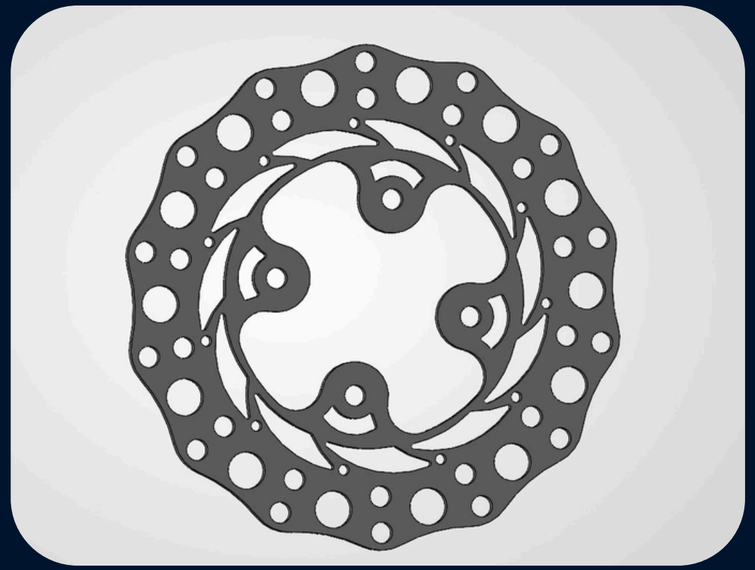
Left Rear Wheel Hub

Brakes

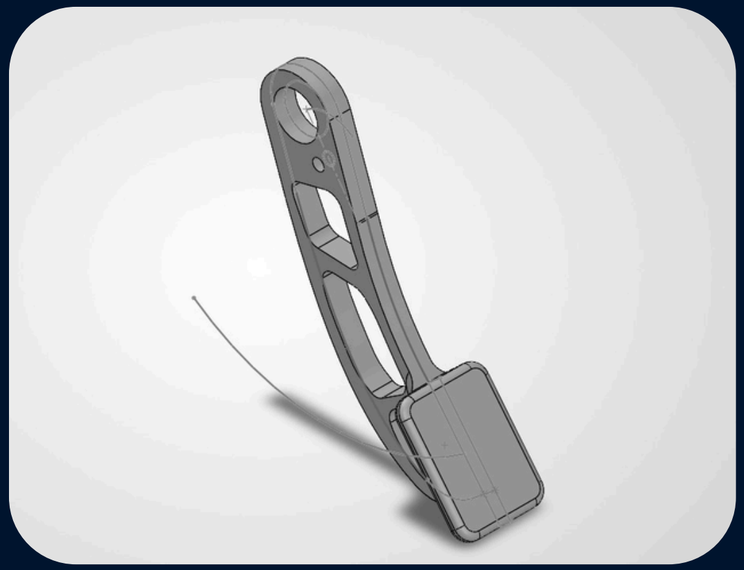
Develops a safe and reliable braking system for maximum braking performance. Ensures compliance with BAJA safety standards through design, testing, and validation.



Brake Circuit Layout



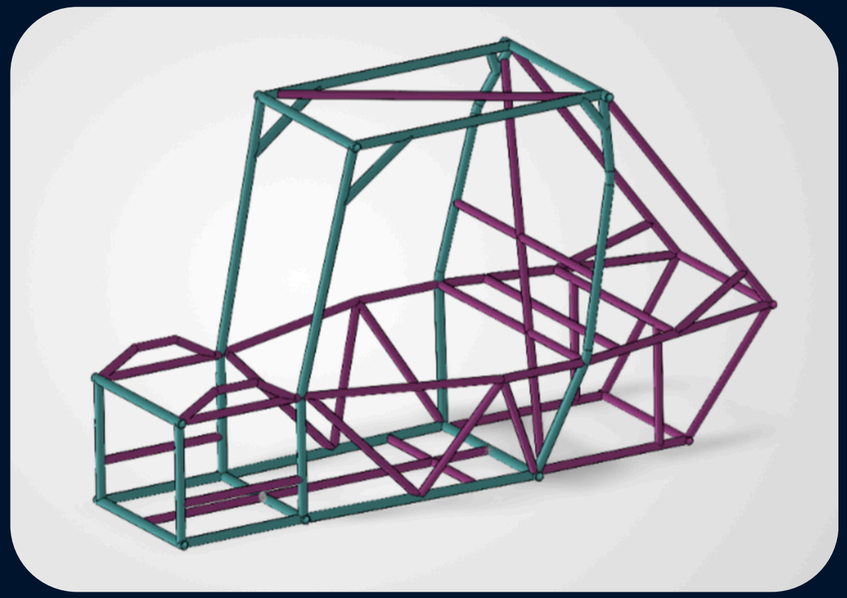
Deformation Analysis of Rotor



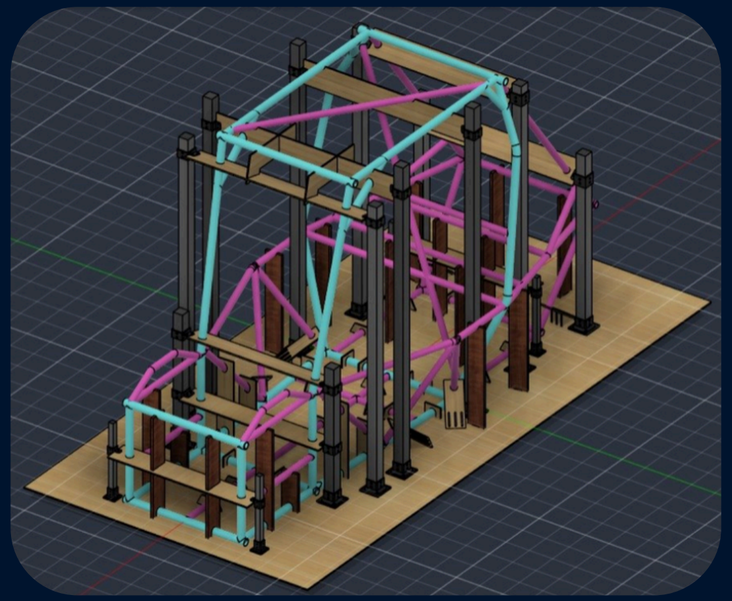
Deformation Analysis of Brake Pedal

Chassis

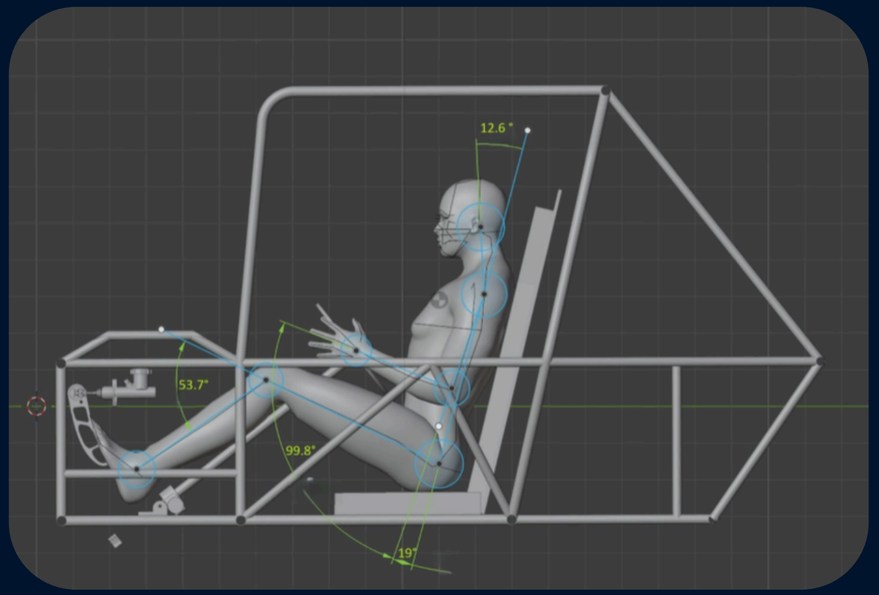
Designs the vehicle rollcage to maximize structural strength while minimizing weight. The design ensures ergonomic driver posture and optimal FOS achieved through successive iterations.



Chassis Isolated View



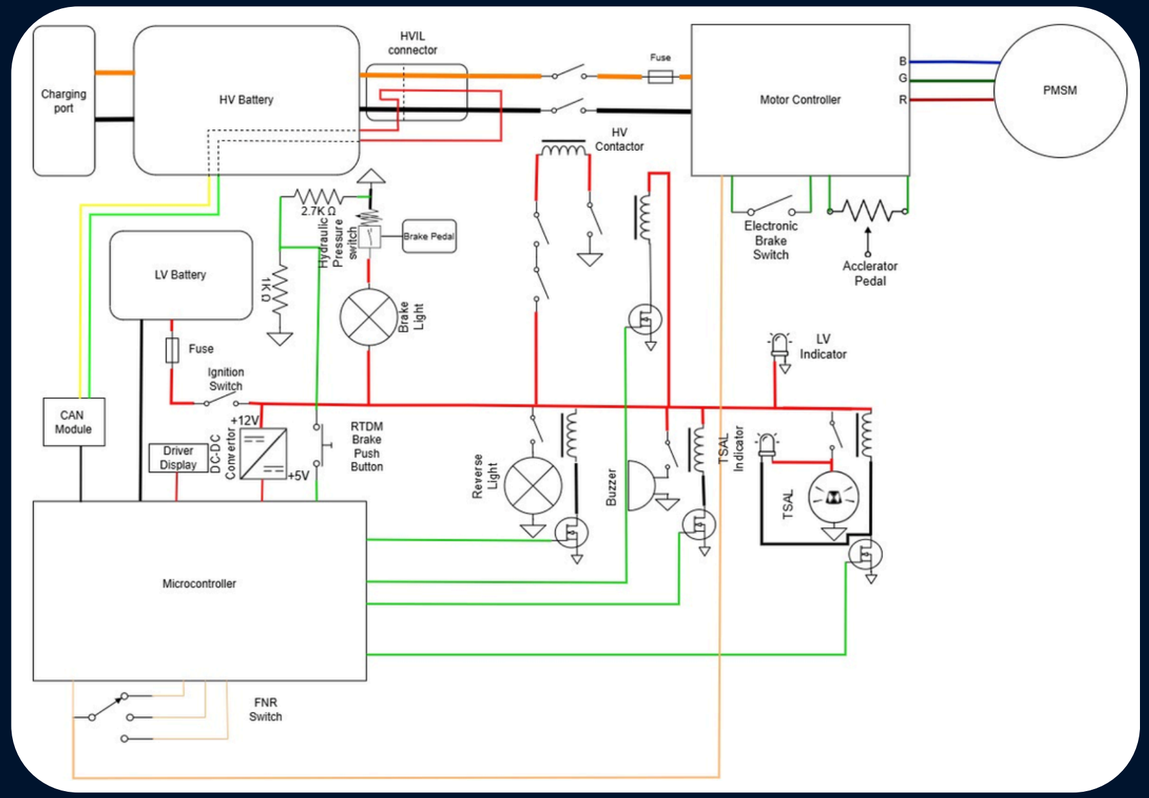
Metal-MDF Jig CAD Visualisation



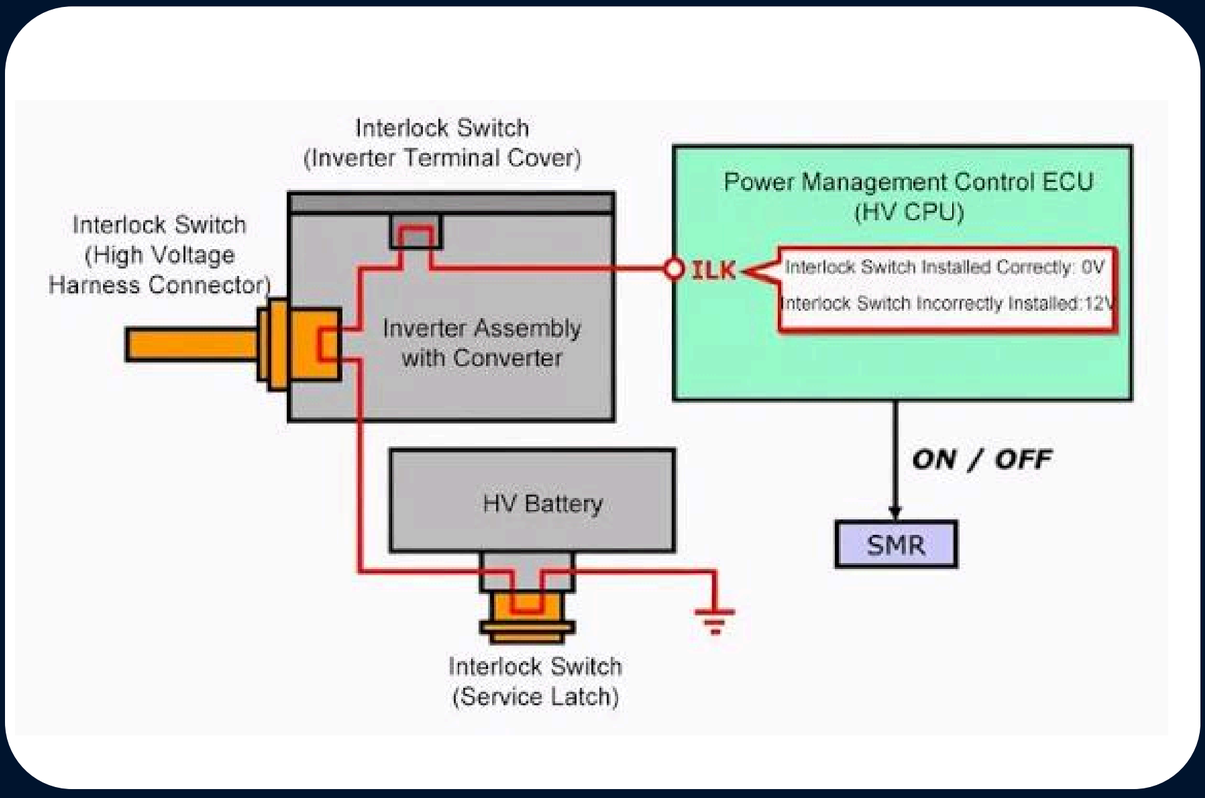
Driver Posture Side View

Electrical Powertrain

Designs and integrates the electrical systems, including the accumulator, motor controller, and wiring. Ensures safe power distribution, battery management, and system reliability.



Tractive System and Complete Circuit Diagram

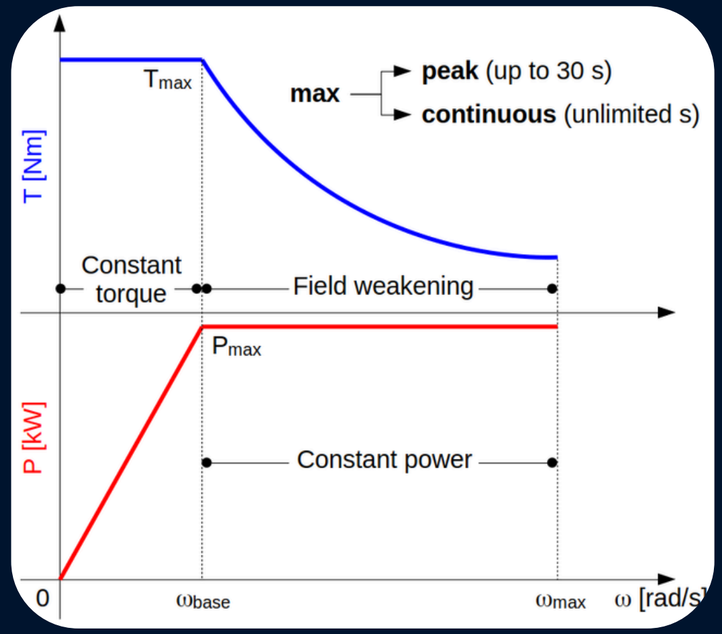


High Voltage Interlock Diagram

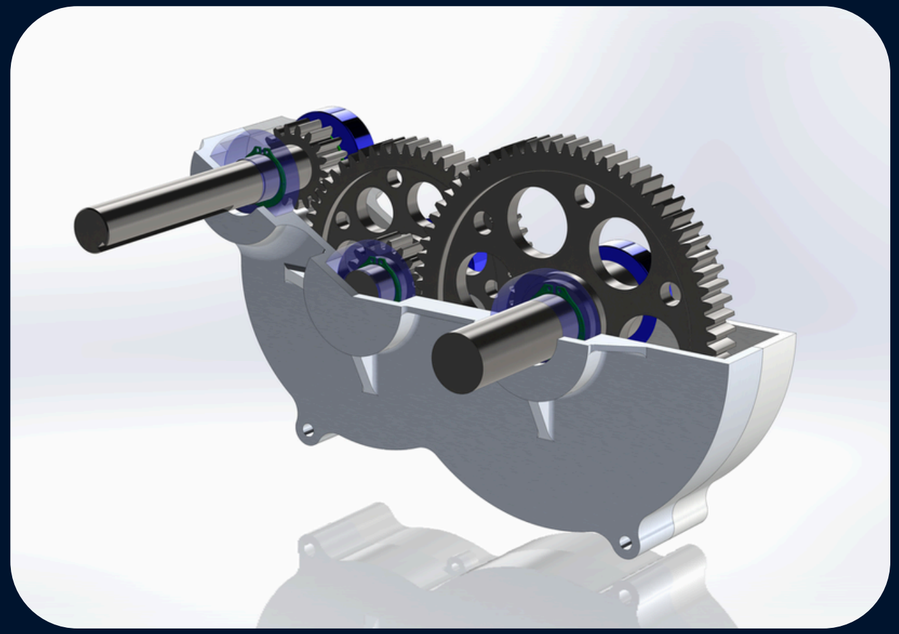
Mechanical Powertrain

Develops the drivetrain, transmission, and power delivery systems. Optimizes efficiency, durability, and performance under demanding off-road conditions.

PMSM Motor



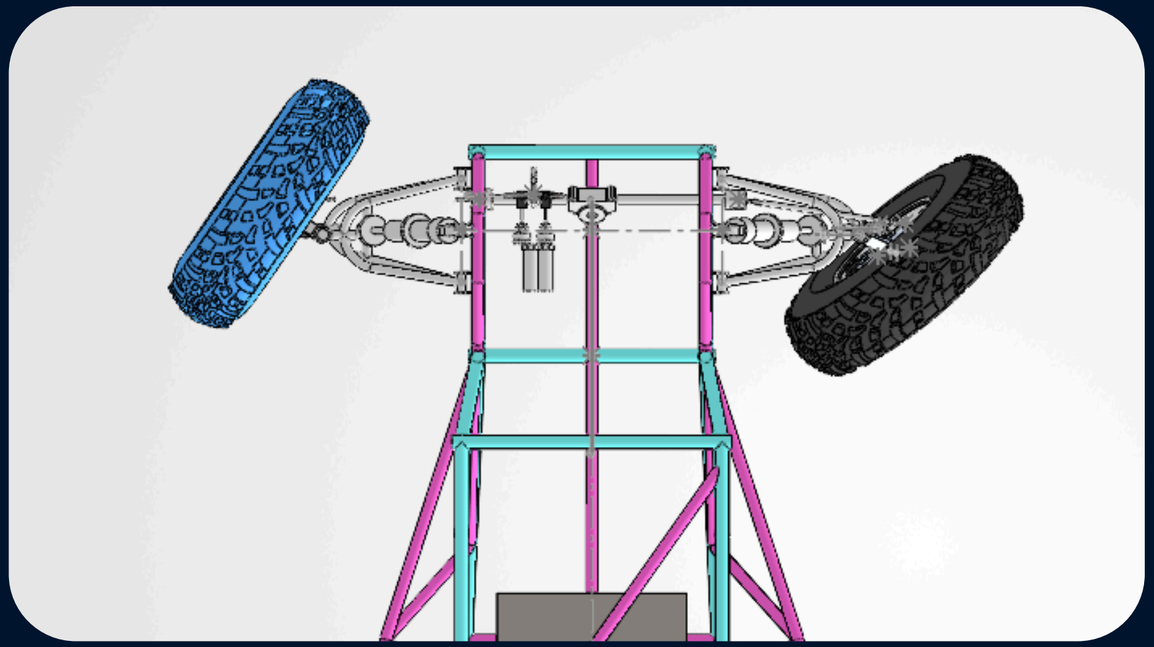
Torque & Power vs RPM curve



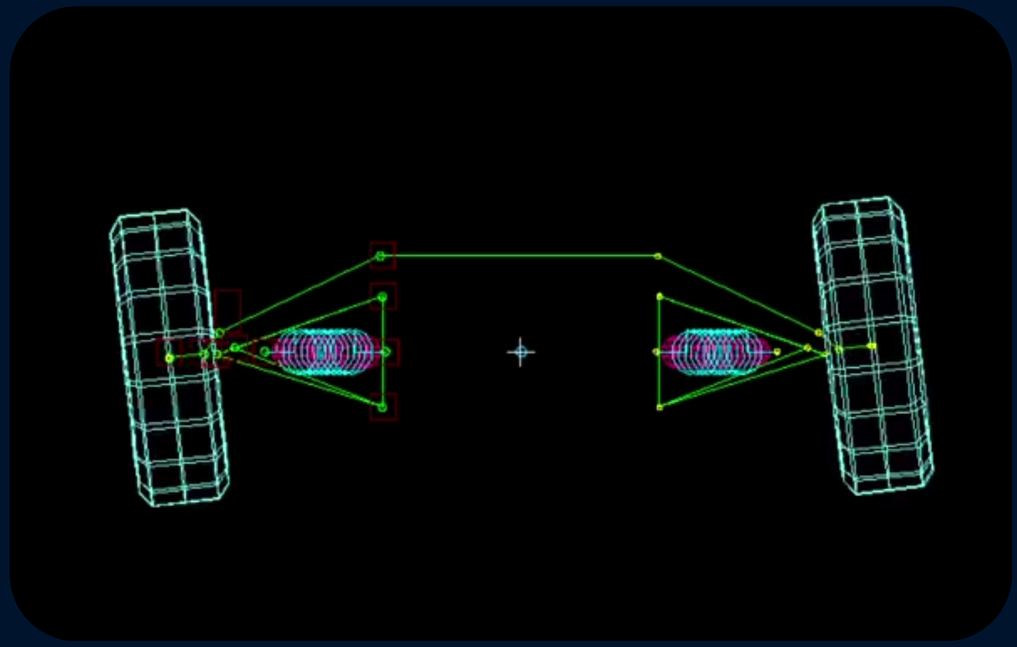
Gear Reducer

Steering

Designs the steering system to provide precise control, responsiveness, and driver confidence. Optimizes steering geometry, ergonomics, and reliability for challenging terrain.



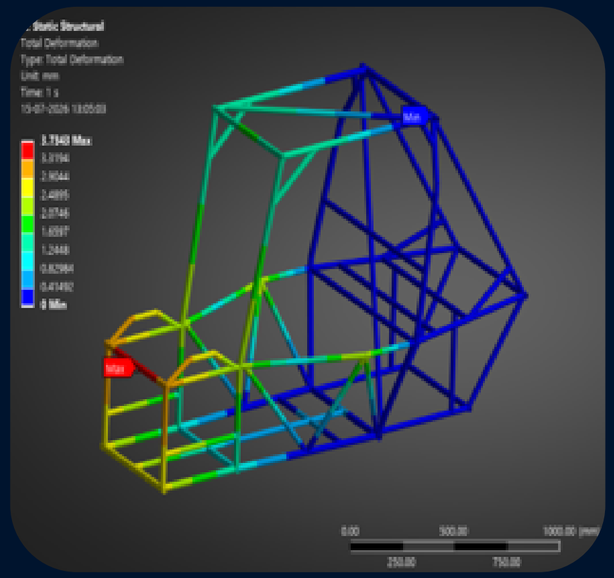
Ackermann Geometry Implemented in our Steering System



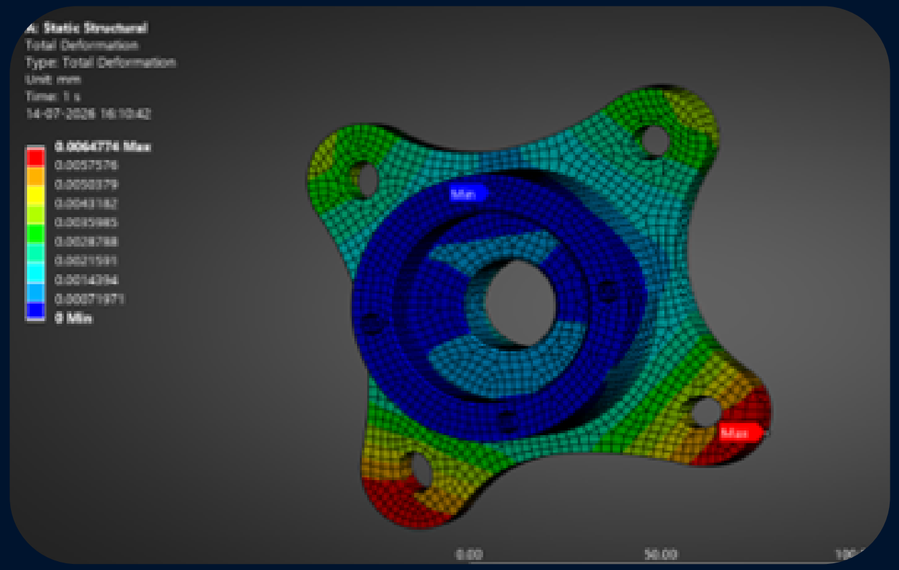
Steering Simulation

CAE Analysis

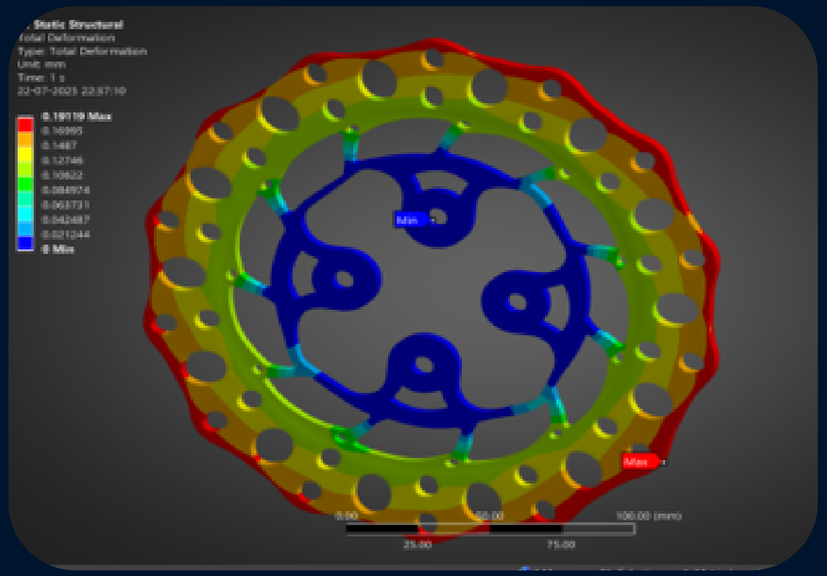
Performs simulations to evaluate structural integrity, performance, and component durability. Uses engineering analysis to validate designs before manufacturing and testing.



Front Collision Stress Analysis



Rear Wheelhub Stress Analysis



Brake Rotor Stress Analysis

Sponsorship

Professional Representation

Sponsor Aquisition

Industry Partnerships

Finance & Procurement

Budget Management

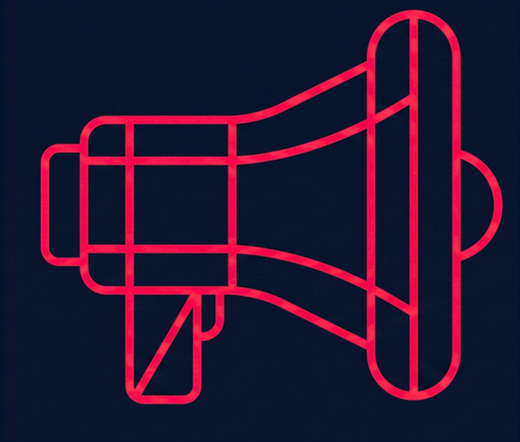
Procurement

Vendor Coordination

Marketing

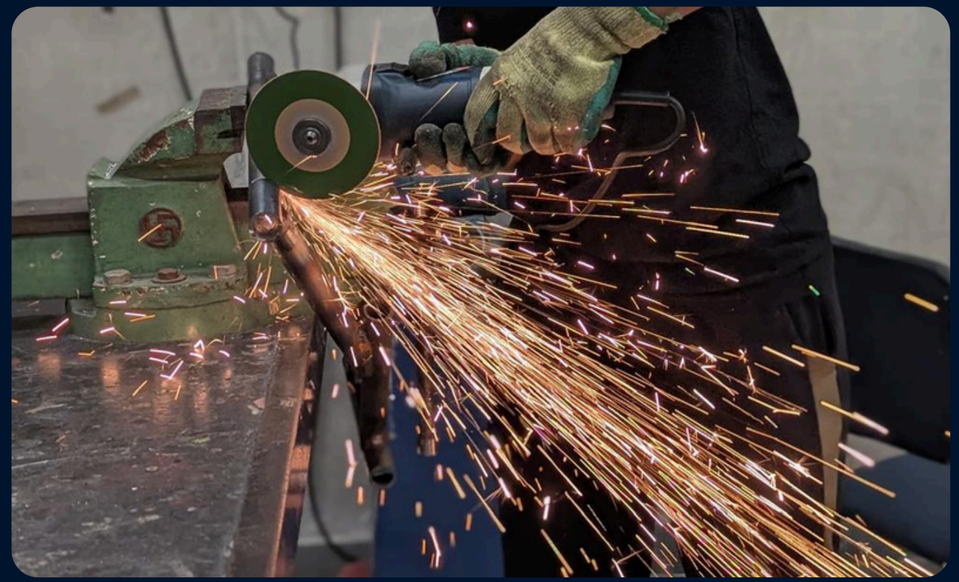
Manage Branding

Create Content



Drive Outreach

Moksha Gallery



Why Join Moksha and What Do We Expect?

- Learn CAD
- Manufacturing trades
- Industry exposure
- Dedication
- Initiative
- Work Ethics
- “Thats all we need, commitment, willingness and dedication”

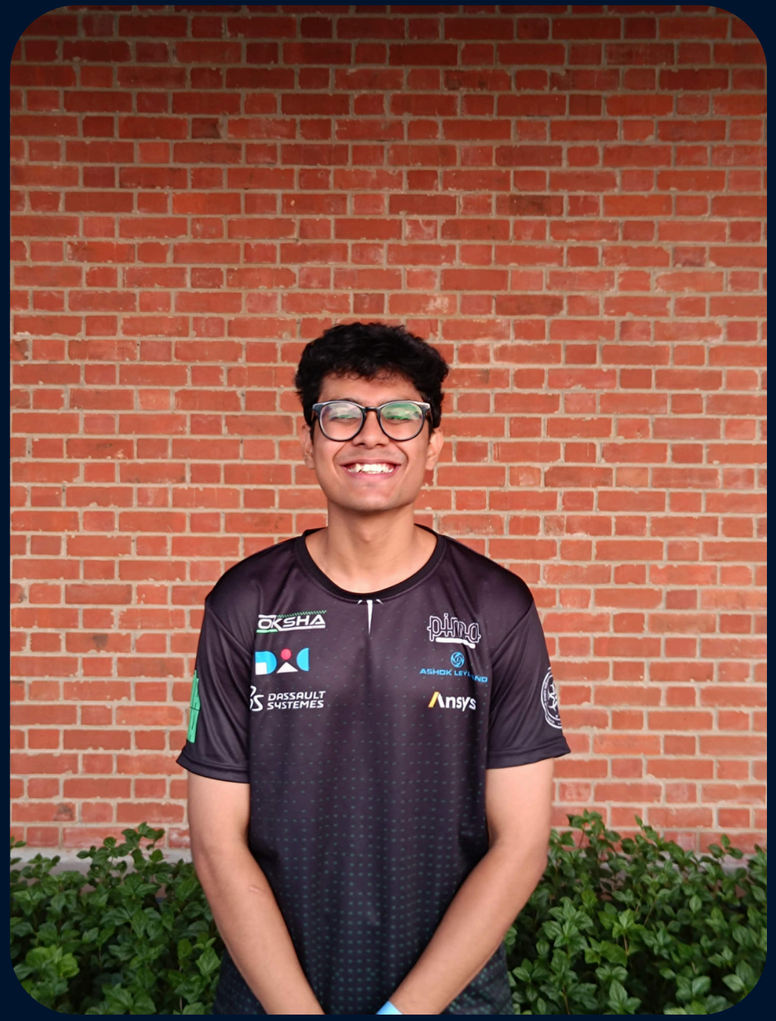
Hear From Those Who Built Moksha

- Aditya Jain - Ex CAE Lead
- B.Tech + M.Tech 4th year in Mechanical Engineering
- Intern @TATA Motors

Hear From Those Who Built Moksha

- Monisha Kavuri - Ex Brakes Lead
- B.Tech 4th year in Mechanical Engineering
- 3 Month Summer Intern @ITC

Hear From Those Who Built Moksha



- Kushagreek Basu - Ex Vice-Captain
- B.Tech 4th year in Mechanical Engineering
- Research Intern @ IISc Bangalore
- Working in Computational & Aerospace Research

Keep yourself posted!

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THANK YOU
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PXGP
Expensify
HAYES